DEWDRIVE · PROJECT PLAN

DewDrive project plan

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
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REVISION HISTORY

Version	Date	Author	Change
0.2	2026-10-25	Amish Chadha	Open items 3 and 4 decided by Amish (2026-10-25)
0.1	2026-10-25	Amish Chadha	First plan (kit 1.8 pipeline), with the reference list of every source cited in the repo

DewDrive project plan

DewDrive is at stage P3 (design outputs) and TRL 3: the constructable design, its calculation note, drawings, priced BOM, build plan, STEM check, concept options, requirements and pipeline documents all exist. The next gate, P4 (verification, TRL 4), is on hold while the portfolio cap is TRL 3. The decisions requested now are the concept selection of DWD-DDR-004 (options A and B within 10 % on the weighted score) and how to recover PR-001 (0.44 L per day against 0.5; DWD-DEC-001, open item 2). Method: STANDARDS sections 19 and 23.

Workstreams

Table 1. Workstreams.

Workstream	Owner (role)	Deliverable	Status
Concept and STEM check	Design lead	DWD-STM-001, DWD-OPT-001, DWD-DDR-004, DWD-DDR-005	Done at P3; STEM check open on 7 rows that need tests, logs or supplier ratings
Requirements	Design lead	DWD-URS-001, DWD-PRS-001, DWD-RTM-001	Done at P3; targets to confirm with the partner
Calculation and CAD	Design lead	DWD-CAL-001 v0.8; `cad/src/model.py` at D3 for options A to C; drawings; STEP	Done at P3
Build plan and BOM	Design lead	DWD-BLD-001; `bom/bom.csv`	Done at P3; quotes for the estimated lines
Regulatory, risk and market	Design lead	DWD-REG-001, DWD-RSK-001, DWD-MKT-001, DWD-FEA-001	First pass done; "to assess" items open
Partner and site	Amish	Agreement with a dryland field station; site humidity and sun record	Not started (partner type decided 2026-10-02)
Verification (P4)	Partner and design lead	Isotherm and absorptance measurements, prototype build, chamber and field tests, water test	On hold (TRL cap 3)

Resources needed

Table 2. Resources (P4 resources are planned work, on hold).

Resource	Deliverable	Effort	Estimated cost (USD)
Prototype parts	One unit to DWD-BLD-001		695 (BOM, estimate)
Sheet-metal and joinery workshop	Box, trays, deck, screens, stand	About 5 days	To assess with the workshop
Coating shop	PTFE coatings of trays, floors, absorbers, screens	One batch	Included in the BOM estimate; to quote
Laboratory (partner or university)	Isotherms of the composite at 20 °C and 80 to 120 °C; absorptance of the absorber over the beads	About 2 weeks	To assess
Water-testing laboratory	Chloride, aluminium and microbiological tests of the condensate	One set per site	To assess
Field station	Site, weather record, 30 days of logged operation	1 month or more	To assess

Stage gates

Table 3. Stages (P4 to P6 on hold while the stage cap is P3).

Stage	Gate criteria	Status
P0 Proposal	PRB with users and stakeholders, STM, OPT, REG, FEA, PLN	Reached (documents added 2026-10-25)
P1 Definition	URS, PRS draft, RSK with hazard analysis, MKT, REG complete	Reached
P2 Design inputs	PRS with targets and verification, CAL, RTM, concept selection DDR, STEM check closed, feasibility closed	Reached, except: concept selection awaiting Amish (DWD-DDR-004); STEM check open on 7 rows that only tests, logs or supplier ratings can close; PR-001 not met on paper
P3 Design outputs	Model, STEP, drawings, priced BOM, build plan, DEC, design FMEA, traceability complete	Reached (TRL 3)
P4 Verification	Test plans and lab reports, verification report, RSK updated	On hold (TRL cap 3)
P5 Validation	Field trials with a partner, human factors, certification status	On hold
P6 Commercialization	Launch plan with a named deployment partner	On hold

Next steps and decision requested

1. **Decisions requested (Amish):** concept selection, DWD-DDR-004 (B 120 and A 117 of 170; option B makes no water at the design point); recovery of PR-001 (DWD-DEC-001, open item 2); fan cut-out and design wind speed (open items 3 and 4) were decided by Amish on 2026-10-25: 65 % RH with restart below 60 %, and a 25 m/s gust until the site value is known.

2. When the TRL cap lifts: measure the composite's isotherms, the absorber's absorptance and the mesh's open share first, because PR-001 is already short on paper.
3. Approach two or three dryland field stations for the first site (risk P3).
4. Get quotes for the estimated BOM lines (coatings, mesh, absorber sheet, glazing grade).

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